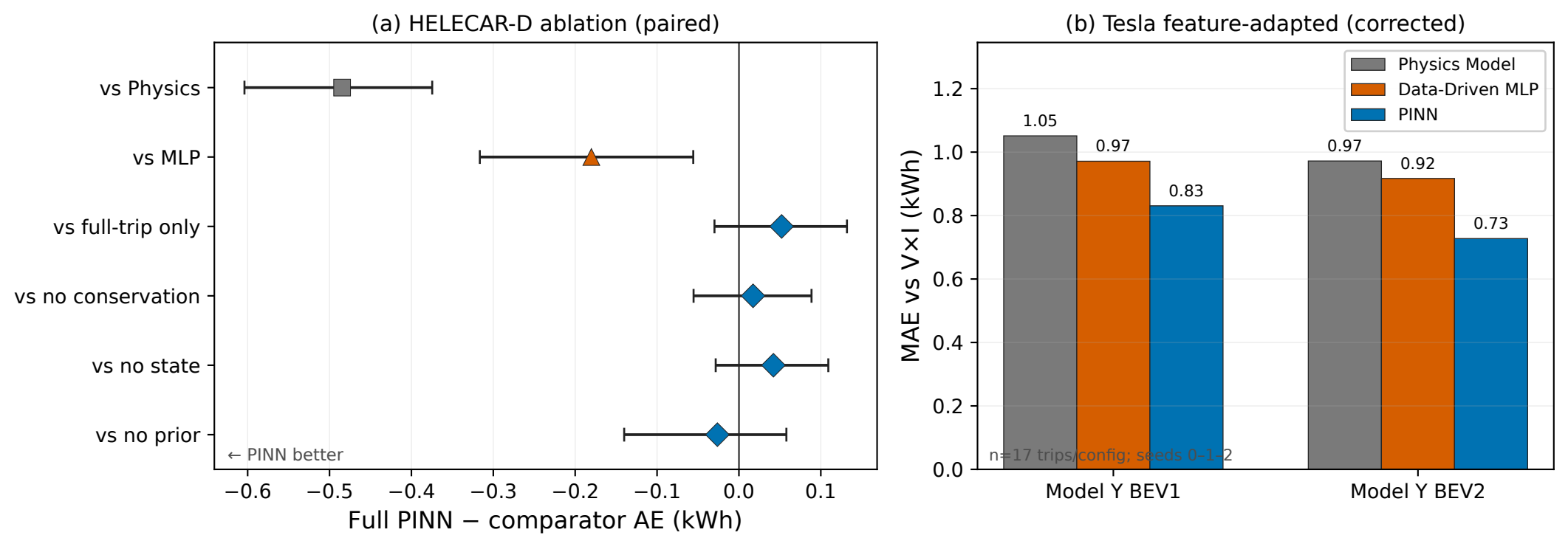
(a) HELECAR-D ablation (paired)
vs Physics
vs MLP
vs full-trip only
vs no conservation
vs no state
vs no prior
← PINN better
−0.6
−0.5
−0.4
−0.3
−0.2
−0.1
0.0
0.1
Full PINN − comparator AE (kWh)
(b) Tesla feature-adapted (corrected)
Physics Model
Data-Driven MLP
PINN
MAE vs V×I (kWh)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.05
0.97
0.83
0.97
0.92
0.73
n=17 trips/config; seeds 0–1–2
Model Y BEV1
Model Y BEV2